

Howard Lin

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science - GPA: 3.83/4.0

June 2026

- **Relevant Coursework:** Operating Systems, Data Structures & Algorithms, Machine Learning/Artificial Intelligence, Computer Security, Database Systems, Object-Oriented Programming

EXPERIENCE

Software Engineer Intern

September 2026 - Present

Transnode AI

Remote

- Integrated Google AdMob using React Native, Expo, and TypeScript, enforcing 100% COPPA compliance (0% ad impressions on kid profiles) and Apple ATT privacy gating
- Identified and fixed a client-side edge case causing cross-device subscription desync and unsafe database record changes by implementing a promote-only sync guard in React Native and verifying status against backend subscription records

Software Engineer Intern

June 2025 – September 2025

MSI (Micro-Star International)

City of Industry, CA

- Streamlined multi-site product review gathering by developing an automated Python and LLM powered summarization + sentiment analysis pipeline, reducing weekly processing times by **67%** across 30+ daily articles
- Automated PDF data extraction by building a Python tool that parses uploaded documents directly into structured CSVs and Google Sheets, reducing manual data entry time by **65%** daily
- Built production frontend interfaces for the MSI Student Sponsorship page in collaboration with UI/UX teams and standardized ad campaign graphics with reusable HTML/CSS templates

PROJECTS

Analytics Platform | JavaScript, Node.js, Express.js, MySQL, Digital Ocean

Live Demo

- Engineered a decoupled web telemetry platform with separate collection and reporting microservices, handling pageviews, Core Web Vitals (LCP, INP, CLS), and JS error grouping with zero client dependencies
- Designed Express REST APIs and MySQL schemas for a dual-storage pipeline (MySQL + JSONL fallback) to ingest event payloads and manage user roles, session logs, and report metadata
- Developed a SPA management dashboard displaying real time analytics, with session authentication, role-based access control (RBAC), interactive session timelines, and headless PDF report generation

Article Summarizer | Python, TypeScript, FastAPI, Supabase, Resend API, Groq API

Live Demo

- Developed an AI-powered web scraping tool utilizing two-level asyncio concurrency to parse and process multiple articles concurrently, and reducing latency and token use by **50%** through a single pass prompt parser
- Built an interactive dashboard featuring a real-time progress bar powered by Server-Sent Events (SSE) to track live scraping and LLM processing steps, and client-side offline HTML report exporting and session caching
- Implemented a multi-stage CI/CD deployment pipeline using GitHub Actions, enforcing automated Python test execution, build artifact generation, and secure file synchronization to a live Linux server
- Designed prompt injection defenses, SSRF protection via cached DNS validation, and PostgREST sanitization

Voice Transcription Tool | Node.js, Express.js, Web Audio API, Whisper large turbo v3

Live Demo

- Built a low-latency voice transcription prototype using Node.js, Express, and Groq Whisper Large v3 Turbo, streaming audio in-memory via the Web Audio API to achieve sub-second latency with zero disk I/O
- Designed an interactive frontend featuring a real-time canvas audio visualizer, cross-browser microphone streaming, word/character counters, and one-click transcript exports

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, C, Java, HTML, CSS, SQL

Frameworks/Libraries: Node.js, Express.js, FastAPI, Flask, Pytest, React/React Native, Expo, Tailwind

Databases/Cloud: MySQL, PostgreSQL, Supabase, Digital Ocean VPS, AWS, Docker

Tools: Git, GitHub, GitHub Actions, Jira, Claude Code, Gemini CLI